nature portfolio

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☐	☒	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	A collection of code used for data collection and analysis is available on Zenodo (https://zenodo.org/record/8052525) as well as on GitHub (https://github.com/DHLocke/HOLC_uneven_biodiv).
Data analysis	All analysis were run using R v. 4.1.1. and the packages tigris v. 1.5, sf v. 1.0.12, rgbif v. 3.7.3.1 and KnowBR 2.0.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

We obtained biodiversity records from the Global Biodiversity Information Facility (https://gbif.org) which contains data from the eBird database (https://ebird.org/). Biodiversity data used in this analysis is publicly available on Zenodo (https://zenodo.org/record/8052525) and was queried as a derived dataset (https://doi.org/10.15468/dd.ha9ksv). Redlining maps were downloaded from the Mapping Inequality Project (https://dsl.richmond.edu/panorama/redlining/).

Climatic predictors used in our analysis are publicly available through the Chelsa Climatology platform (https://chelsa-climate.org/) and remote sensing data is available on Google Earth Engine (https://earthengine.google.com) and the PAD-US dataset can be downloaded from the United States Geological Survey website (https://www.usgs.gov/)

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	Not applicable
Population characteristics	Not applicable
Recruitment	Not applicable
Ethics oversight	Not applicable

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Historic segregation and inequality are critical to understanding modern environmental conditions. Race-based zoning policies, such as redlining in the United States during the 1930s, are associated with racial inequity and adverse multigenerational socioeconomic levels in income and education, and disparate environmental characteristics including tree canopy cover across urban neighborhoods. We quantify the association between redlining and bird biodiversity sampling density and completeness - two critical metrics of biodiversity knowledge - across 195 cities in the United States. We show that historically redlined neighborhoods remain the most under-sampled urban areas for bird biodiversity today, potentially impacting conservation priorities and propagating urban environmental inequities. The disparity in sampling across redlined neighborhoods grades increased by approximately 40% over the past 20 years. We identify specific areas critical for reducing uncertainty and increasing equity of sampling of biodiversity in urban areas. Our findings highlight how human behavior and past social, economic and political conditions segregate not just our built environment, but may also leave a lasting mark also the on the digital information we have about urban biodiversity.
Research sample	We obtained a total of 12,297,506 georeferenced biodiversity records for bird species (class Aves) occurring within 9,851 HOLC-defined neighborhoods in 195 cities, from the Global Biodiversity Information Facility (https://gbif.org), which also contains records from the eBird database 58. We used the rgbif package v. 3.7.3.1 59 to query and download bird records that had geographic coordinates, and were collected after 1932 (as the Home Owners Loan Corporation began categorizing neighborhoods in 1933) up to October 2022. We included records defined as 'observation', 'living specimen', 'human observation', 'preserved specimen' and 'machine observation'.
Sampling strategy	As we were interested in the sampling of bird biodiversity and the amount of records per km2, we downloaded all bird records occurring within our defined HOLC neighborhoods in 195 cities collected between 1933 and 2022.
Data collection	We obtained bird biodiversity records from the Global Biodiversity Information Facility (GBIF) using the 'rgbif' package version 3.7.3.1 in R v. 4.1.1.
Timing and spatial scale	We queried and downloaded bird records that had geographic coordinates intersecting with our HOLC polygons and were collected from 1933 up to 2022.
Data exclusions	After downloading biodiversity records defined as 'observation', 'living specimen', 'human observation', 'machine observation' from the Global Biodiversity Information Facility (GBIF), we did not exclude any datasets
Reproducibility	Data and code used in this analysis is publicly available on Zenodo repository (https://zenodo.org/record/8052525) as well as on GitHub (https://github.com/DHLocke/HOLC_uneven_biodiv)
Randomization	As we were interested in the sampling of biodiversity records, we did not randomize any samples
Blinding	Blinding was not used for data acquisition in this manuscript as we downloaded more than a dozen million bird records for our study across and within 195 metropolitan areas in the United States across 38 states

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging